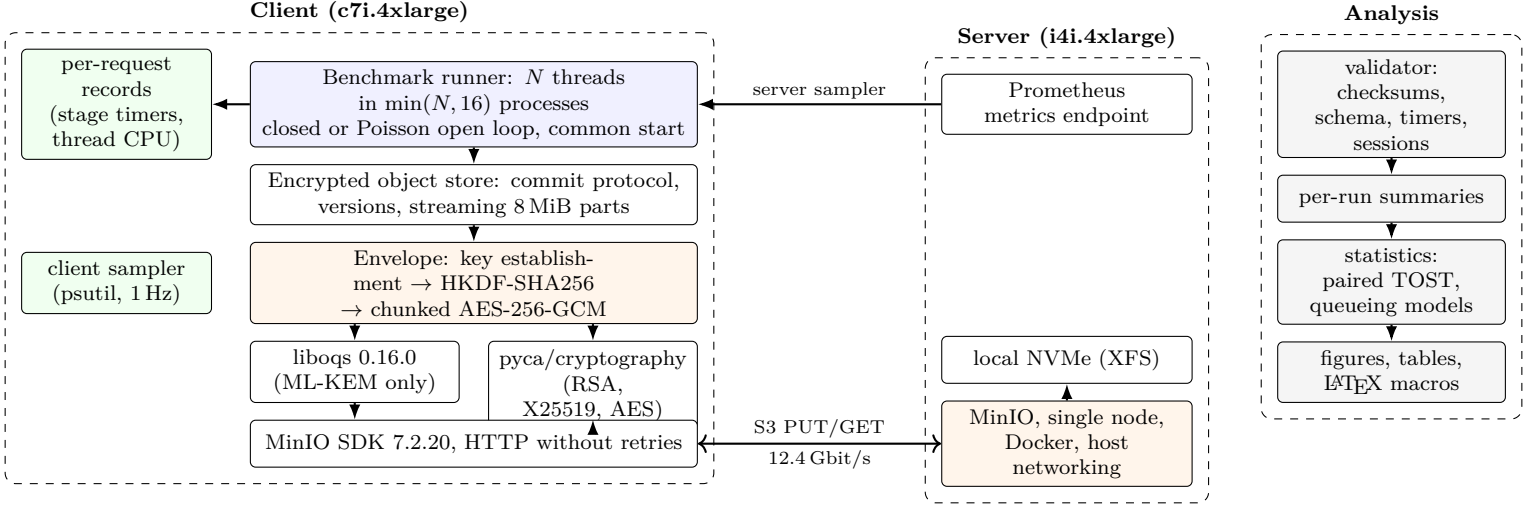
Client (c7i.4xlarge)
per-request records (stage timers, thread CPU)
Benchmark runner: N threads in $\min(N, 16)$ processes closed or Poisson open loop, common start
Encrypted object store: commit protocol, versions, streaming 8 MiB parts
client sampler (psutil, 1 Hz)
Envelope: key establishment → HKDF-SHA256 → chunked AES-256-GCM
liboqs 0.16.0 (ML-KEM only)
pyca/cryptography (RSA, X25519, AES)
MinIO SDK 7.2.20, HTTP without retries
server sampler
S3 PUT/GET
12.4 Gbit/s
Server (i4i.4xlarge)
Prometheus metrics endpoint
local NVMe (XFS)
MinIO, single node, Docker, host networking
Analysis
validator: checksums, schema, timers, sessions
per-run summaries
statistics: paired TOST, queueing models
figures, tables, LaTeX macros